JobPilot Technical Brief

BAX-423 Final Project Option B: Smart Job Matcher

1. System Overview

JobPilot is a runnable Streamlit application that recommends jobs from a 20,000-posting multi-country real snapshot extracted from the professor-recommended Techmap/Kaggle jobs dump. The system supports the six required capabilities: ingestion, profile intake, dense retrieval, multi-stage ranking, adaptive feedback learning, top-job download, explanations, analytics, and tailored resume generation.

Architecture: Techmap/Kaggle zip stream parser -> normalized multi-country job snapshot -> simulated stream replay -> SQLite dedup store -> text normalization and skill extraction -> TF-IDF/SVD dense embeddings -> country-aware candidate filtering -> nearest candidate retrieval -> multi-stage ranker -> Streamlit UI with explain, feedback, analytics, CSV export, and resume draft.

Layer	Implementation	Production Upgrade
Data	20,000-row real Kaggle/Techmap sample across countries and fields, replayed in batches into SQLite with job_id primary-key dedup.	Add current API fetchers -> Pub/Sub/Kafka -> Cloud SQL/BigQuery.
Model	Dense vectors from TF-IDF + SVD; country filter before cosine retrieval; ranking combines similarity, skill overlap, role, salary/location, and dealbreake	Vector DB, scheduled refreshes, evaluation dashboard, drift checks.
Serving	Streamlit app with persona controls, recommendation cards, explanations, feedback, and downloads.	Cloud Run or Streamlit Cloud with auth and managed secrets.
Monitoring	Persona pass table and benchmark tab visible in app.	Logging, latency/error metrics, ranking-quality alerts.

2. BAX-423 Techniques and Benchmarks

Technique A: streaming-style ingestion and deduplication. The project streams the large Techmap/Kaggle zip without extracting the full 50GB JSON file, samples 20,000 real postings across multiple countries and fields, writes a normalized CSV, then replays postings in batches into SQLite and skips duplicate job_id values. This mirrors the lecture's prototype-to-production data layer: the app does not depend on live API calls during demo, but the boundary is ready for a real stream and current API refresh.

Technique B: embedding-based retrieval plus multi-stage ranking. Jobs and profiles are represented as dense vectors. The system first filters to the persona's preferred countries, retrieves candidates, then adjusts for skill overlap, target-role alignment, salary/location preferences, user feedback, and dealbreaker penalties.

Persona	Keyword Skill Overlap	Multi-stage Skill Overlap	Keyword Violations	Multi-stage Violations
Aisha	1.0	0.547	5	0
Marcus	1.0	0.556	5	0
Priya	1.0	0.723	3	0
Kenji	1.0	0.429	0	0
Sofia	1.0	0.727	1	0
Lukas	1.0	0.422	3	0
Mei	1.0	0.353	1	0
Arjun	1.0	0.472	1	0
Nora	1.0	0.375	3	0
Elena	1.0	0.833	0	0
David	1.0	0.773	1	0
Leila	1.0	1.0	1	0
Maya	1.0	0.758	1	0
Omar	1.0	1.0	1	0

Interpretation: keyword matching can produce high overlap but ignores country and hard constraints. The multi-stage pipeline reduced top-10 hard dealbreaker violations to 0 across the required and extended personas while preserving role relevance.

3. Core Capabilities

Requirement	How JobPilot Implements It
Job ingestion & streaming	CSV snapshot is replayed into SQLite in batches; duplicate job_id inserts are rejected. The UI exposes the replay result.
Profile intake & skill matching	Sidebar accepts persona, resume/profile text, skills, target roles, preferred countries, salary, and dealbreakers.
Embedding retrieval	Profile and job text are embedded into dense vectors using TF-IDF/SVD. Cosine similarity retrieves candidate jobs.
Multi-stage ranking	Preferred countries are applied before retrieval; final score combines embedding similarity, skill overlap, role alignment, salary/locatio
Adaptive learning	Accept/reject/skip modifies skill-level preference weights and immediately changes ranking in the session.
Download top jobs	A CSV download button exports job_id, title, company, country, location, salary, apply link, and description for current top matches.
Explain & resume	Each recommendation shows score drivers and cautions; every job has a tailored resume draft and text download.
Batch analytics	Analytics tab shows top skills, salary by role family, and demand by country/location.

4. Persona Test Results

Persona	Countries	Result	Mean Score
Aisha	United States, Canada	PASS	0.754
Marcus	United States, Canada, United Kingdom	PASS	0.774
Priya	United States	PASS	0.744
Kenji	United States, Canada	PASS	0.691
Sofia	United Kingdom, Ireland	PASS	0.752
Lukas	Germany, Switzerland, Netherlands	PASS	0.634
Mei	Singapore, Hong Kong, Australia	PASS	0.653
Arjun	India, Singapore	PASS	0.631
Nora	Canada	PASS	0.534
Elena	United Kingdom, Ireland	PASS	0.783
David	Germany, Switzerland	PASS	0.59
Leila	United Arab Emirates, United Kingdom	PASS	0.654
Maya	India, United Kingdom, Vietnam	PASS	0.713
Omar	Canada, United States	PASS	0.895

All four required course personas plus ten additional personas pass in the current snapshot. The extended set covers analytics/ML, marketing, finance, healthcare operations, HR/recruiting, and sales across North America, Europe, and Asia-Pacific.

5. Limitations and Production Path

- The included data is a real September 2021 Techmap/Kaggle snapshot, not a live Adzuna/JSearch feed. A production version should refresh postings daily and store API provenance.

- The resume generator is template-based and should be guarded by honesty checks, ATS review, and optional LLM prompt versioning before real use.

- Feedback learning is session-level. A deployed app should persist user-specific feedback, evaluate drift, and avoid overfitting to a few clicks.

- Current hosting is local/deploy-ready. To satisfy live hosting, push to GitHub and deploy `code/app.py` to Streamlit Cloud or Cloud Run.